

TETHER: Supplementary Material

Anonymous Authors

Abstract—This document reports measurements that did not fit the page limit. It adds no claim to the main paper. Two of its sections (Sec. III and Sec. VII) contain results that *qualify* the main claims rather than strengthen them, and they are included for that reason.

I. COMPLETE MATCHED-BASELINE CLOSURE

Table I lists all fifteen policies of the D2 closure. The main paper reports a representative subset. Every row shares the bounded state machine, the raw re-anchor schedule, the SEA-RAFT alignment, the support definition and the fusion form; only the blending weight differs, and all 225 reports (15 methods $\times$ 5 backbones $\times$ 3 sequences) come from a single run under one frozen protocol.

TABLE I

FULL SCARED-C D2 MATCHED TEMPORAL-POLICY CLOSURE. RAW EPE IS 0.276202. POSITIVE REDUCTION DENOTES IMPROVEMENT.

Method	EPE	Reduction
TETHER $H = 6$	0.264656	4.18%
TETHER $H = 8$	0.264912	4.09%
TETHER $H = 4$	0.265028	4.05%
TETHER $H = 2$	0.267265	3.24%
TETHER $H = 1$	0.269001	2.61%
Fixed $w = 0.5$	0.274562	0.59%
EMA2	0.274562	0.59%
Fixed $w = 0.3$	0.274852	0.49%
Fixed $w = 0.2$	0.275229	0.35%
EMA3	0.275386	0.30%
Fixed $w = 0.1$	0.275684	0.19%
Warped raw previous	0.276298	−0.03%
FB confidence weight	0.282182	−2.17%
Warped recurrent $w = 1$	0.286035	−3.56%
Continuous recurrence	0.449349	−62.69%
Oracle raw/memory	0.242223	12.30%

a) *An identity worth stating.*: EMA2 and Fixed $w = 0.5$ are the same policy under two names, and EMA3 is $w = 2/3$: under a recurrent state a fixed convex weight *is* an exponential moving average. The baseline family is therefore five fixed weights $w \in \{0.1, 0.2, 0.3, 0.5, 2/3\}$ over the shared bounded state, not two independent families. We state this because two nominally different baselines producing an identical number otherwise reads as a bug.

II. THE ORACLE CEILING IS NOT THE ONE WE REPORT AGAINST

The oracle of the main paper selects, per pixel, the better of raw disparity and aligned memory. It is therefore the ceiling of $w \in \{0, 1\}$, whereas TETHER acts on $w \in [0, 1]$, whose

ceiling is the continuous oracle $w^* = \text{clip}((g - d^{\text{raw}})/(\tilde{m} - d^{\text{raw}}), 0, 1)$.

TABLE II

BINARY VERSUS CONTINUOUS ORACLE ON D2, PER BACKBONE.

Backbone	raw	TETHER	$\{0, 1\}$	$[0, 1]$
S ² M ² -S	.2949	.2850	.2634	.2610
RAFT-Stereo	.3022	.2914	.2687	.2658
Stereo Anywhere	.3189	.2964	.2747	.2720
CREStereo	.2943	.2832	.2610	.2582
Fast-FoundationStereo	.3037	.2930	.2715	.2688

The correction is real but small. Only 6.1% of D2 pixels have the ground truth bracketed by the two candidates (11–14% on D7), so the two ceilings nearly coincide and the recovered fraction falls from 32.9% to 30.1%. We report it because the published figure is measured against a ceiling the method cannot actually be limited by, not because it changes the conclusion.

III. MOTION-COMPENSATED MULTI-HORIZON FIDELITY

TEPE_{corr} and OPW require a correspondence field, supplied here from the same SEA-RAFT flows used for warping. Table III gives the relative change after refinement.

On held-out D7 both improve in all 20/20 backbone–horizon cases. On development D2 they disagree: OPW improves everywhere while TEPE_{corr} degrades everywhere.

This rules out a convenient explanation. One might attribute a rising temporal-error metric to a grid metric penalising legitimate scene motion, but TEPE_{corr} *is* motion compensated and still degrades on D2. The two ask different questions: OPW asks whether consecutive predictions agree with each other, TEPE_{corr} whether the predicted temporal change matches the true one. On the split used to select the operating point, the module therefore becomes more self-consistent while tracking the true change less faithfully. We report this against our own interest.

IV. FORENSICS OF THE SUPPORT VIOLATION

The main paper states that an unmasked output lets zero-padded out-of-bounds memory become an invalid prediction. The supporting measurement is below: every offending pixel has an aligned memory of *exactly* zero, and the offenders arrive in bursts no longer than the state lifetime, so bounded recurrence contains the damage it cannot prevent.

Because an invalid prediction on valid support is penalised rather than discarded, at 10⁴ mm per pixel, a few hundred pixels out of 38.8 million decide the external-domain metric-depth verdict.

TABLE III

TEPE_{corr} (TOP) AND OPW (BOTTOM), RELATIVE CHANGE AFTER REFINEMENT. NEGATIVE IS BETTER.

Split	Backbone	$k=1$	$k=2$	$k=4$	$k=8$
TEPE _{corr}					
D2	S ² M ² -S	+24.5%	+40.1%	+12.9%	+19.7%
D2	RAFT-Stereo	+24.8%	+41.1%	+16.6%	+23.8%
D2	Stereo Anywhere	+14.5%	+31.1%	+8.5%	+16.4%
D2	CREStereo	+22.2%	+39.7%	+15.3%	+24.3%
D2	Fast-FoundationStereo	+35.9%	+52.3%	+21.1%	+27.8%
D7	S ² M ² -S	-23.7%	-21.0%	-17.4%	-12.7%
D7	RAFT-Stereo	-12.2%	-6.1%	-5.6%	-5.7%
D7	Stereo Anywhere	-10.3%	-4.8%	-5.3%	-5.7%
D7	CREStereo	-13.6%	-6.8%	-6.7%	-6.4%
D7	Fast-FoundationStereo	-10.9%	-4.9%	-5.6%	-6.2%
OPW					
D2	S ² M ² -S	-8.7%	-3.5%	-1.8%	-1.2%
D2	RAFT-Stereo	-8.5%	-3.4%	-1.7%	-1.1%
D2	Stereo Anywhere	-12.9%	-5.8%	-3.0%	-1.8%
D2	CREStereo	-9.5%	-3.8%	-1.8%	-1.2%
D2	Fast-FoundationStereo	-7.3%	-3.1%	-1.6%	-1.2%
D7	S ² M ² -S	-29.8%	-22.8%	-13.5%	-7.6%
D7	RAFT-Stereo	-18.5%	-10.2%	-5.1%	-3.4%
D7	Stereo Anywhere	-18.1%	-9.9%	-5.1%	-3.4%
D7	CREStereo	-20.3%	-11.1%	-5.9%	-3.8%
D7	Fast-FoundationStereo	-18.5%	-10.0%	-5.2%	-3.6%

TABLE IV

INVALID REFINED PIXELS UNDER THE UNMASKED OUTPUT, AND THE STATE AGE AT WHICH THEY OCCUR.

	Vid10_Liver_Med	Vid11_Liver_High
Support pixels	38,828,160	38,439,360
Invalid (unmasked)	300	265
Invalid (support-confined)	0	0
Frames affected	10	3
Aligned memory, min/max	0.0 / 0.0	0.0 / 0.0
Non-finite memory	0	0
State age 1/2/3/4	10/103/135/52	0/182/72/1

V. PER-RECORDING DRENDS TRANSFER

All 60 recording–backbone–metric cells improve. The per-recording pattern is set by the recording and not by the estimator: on Vid10 the EPE change is -5.7% , -5.2% and -5.0% for the seen and the two unseen backbones. That alignment, rather than the pooled mean, is what supports backbone independence.

VI. STAGE-WISE RUNTIME

NVIDIA H100 PCIe, PyTorch 2.5.1+cu121, 144×180 evidence grid, 200 timed iterations after 30 warmup iterations.

Two observations. The 177k-parameter head costs 1.03 ms, 3% of the overhead: “compact head” is true but not the operative cost, since 79% is the two frozen flow passes. And the reverse pass costs 13.19 ms while feeding only the forward–backward confidence cue, whose direct use as a fusion weight worsens EPE by 2.17% (Table I); removing it would cut overhead by 39% and is the most promising efficiency ablation we have not run.

TABLE V

ZERO-SHOT DRENDS, RELATIVE CHANGE AFTER REFINEMENT, ALL METRICS AND ALL FIVE RECORDINGS. NEGATIVE IS BETTER.

Backbone	Recording	EPE	Bad3	d.MAE	d.RMSE
RAFT-Stereo (seen)	Vid10	-5.7%	-11.3%	-3.8%	-3.4%
	Vid11	-3.4%	-21.9%	-2.5%	-1.5%
	Vid12	-8.4%	-22.7%	-6.7%	-6.0%
	Vid13	-4.3%	-14.2%	-3.2%	-3.0%
CREStereo (unseen)	Vid10	-5.2%	-8.7%	-3.4%	-3.2%
	Vid11	-3.7%	-22.8%	-2.7%	-1.8%
	Vid12	-6.3%	-18.5%	-4.8%	-4.3%
	Vid13	-5.2%	-10.4%	-3.6%	-3.3%
Fast-Found. (unseen)	Vid14	-5.6%	-26.5%	-4.4%	-4.1%
	Vid10	-5.0%	-7.3%	-3.0%	-2.9%
	Vid11	-3.2%	-20.4%	-2.3%	-1.4%
	Vid12	-5.3%	-10.3%	-4.4%	-3.6%
	Vid13	-4.3%	-10.9%	-3.0%	-2.9%
	Vid14	-4.7%	-27.4%	-3.8%	-3.5%

TABLE VI

PER-STAGE TEMPORAL OVERHEAD.

Stage	median (ms)	p95 (ms)
SEA-RAFT forward	13.222	13.289
SEA-RAFT reverse	13.191	13.268
Alignment	1.038	1.057
142-channel evidence	5.122	5.156
Fusion head	1.027	1.043
Total overhead	33.601	
Peak VRAM	124.2 MB	

VII. THREE-DIMENSIONAL SURFACE CONSISTENCY

SCARED-C provides no per-frame camera pose, so global reconstruction drift cannot be measured. We instead back-project each frame’s disparity to a metric cloud and compare it against the previous frame’s cloud pulled through the same SEA-RAFT correspondence: frame-to-frame surface stability, not reconstruction drift.

Excess over the GT floor falls on four backbones of five, for instance $0.0326 \rightarrow 0.0276$ mm on RAFT-Stereo, and rises slightly on Fast-FoundationStereo. The effect is real but of the order of 0.005 mm, which is why it does not appear in the paper.

The normal metric carries a caveat that removes its floor entirely: GT normals are *noisier* than every prediction (5.86° against $3.74\text{--}5.60^\circ$), because the pseudo-ground-truth is sparse and central differences amplify it. Predictions smoother than the reference cannot be scored against that reference, so the normal column should be read as a relative comparison between raw and refined only.

Further assumptions a reader may wish to challenge: the principal point is taken at the image centre with $f_y = f_x$, since the SCARED-C manifest carries only f_x and the baseline; flow correspondence stands in for true scene-point correspondence, gated by forward–backward consistency at library defaults; and both clouds live in their own camera

TABLE VII

FRAME-TO-FRAME 3D SURFACE STABILITY ON D2, POOLED MEANS.
GT IS THE FLOOR SET BY REAL SCENE AND CAMERA MOTION.

Backbone	raw	refined	GT
<i>point-to-point distance [mm]</i>			
S ² M ² -S	0.5016	0.4978	0.4758
RAFT-Stereo	0.5085	0.5034	0.4758
Stereo Anywhere	0.5060	0.4996	0.4758
CREStereo	0.5065	0.5004	0.4758
Fast-FoundationStereo	0.4890	0.4902	0.4758
<i>surface-normal angle [deg]</i>			
S ² M ² -S	4.4878	4.3241	5.8586
RAFT-Stereo	5.1443	4.7297	5.8586
Stereo Anywhere	5.4499	4.8516	5.8586
CREStereo	5.6006	4.9958	5.8586
Fast-FoundationStereo	3.7381	3.8742	5.8586

frames, so the measure mixes surface change with rigid camera motion, a component the GT floor absorbs but does not isolate.

VIII. LARGER TRAINING DOES NOT MOVE THE OPERATING POINT

An augmented-training run keeps the architecture and enlarges the data.

TABLE VIII

HELD-OUT D7, CANONICAL VERSUS AUGMENTED TRAINING.

Metric	Canonical	Augmented
Mean EPE	.4693	.4706
HUR	.3965	.4457
H^+	.0148	.0235
B^+	.0376	.0450
BUR	.4951	.5060
Degraded-frame fraction	.2642	.3341
Worst frame degradation	.1063	.3958

The augmented model performs more beneficial updates and substantially more harmful ones. Scale shifts intervention aggressiveness without improving held-out risk–gain balance.

IX. EXTERNAL STABILISER CONTEXT

BiDAStabilizer [1] is the closest plug-in stabiliser in our comparison set. Its full formulation is non-causal, using future-frame bidirectional propagation. On the full SCARED-C D2 sequence set our independent reproduction moves pixel-weighted EPE from 0.302862 to 0.302109.

We deliberately draw no comparison from this. The two evaluation paths use independently validated but not pixel-identical support contracts, and the raw EPE differs between them (0.3029 against 0.2762), so the absolute values are not subtractable. A common-support table is the correct way to make this comparison and we have not produced one.

X. A METRIC FIELD WE DO NOT REPORT

FrameDegradation’s CatastrophicFraction was numerically identical to PositiveFraction on every inspected row. The cause is definitional: the two are $\Pr(D_t > \tau_{\text{cat}})$ and $\Pr(D_t > 0)$, and the catastrophic threshold defaults to $\tau_{\text{cat}} = 0$, which the evaluators never override. At that setting they coincide by construction. We therefore report only the degraded-frame fraction, defined explicitly as $\Pr(D_t > 0)$.

REFERENCES

- [1] J. Jing, Y. Mao, A. Qiu, and K. Mikolajczyk, “Match Stereo Videos via Bidirectional Alignment,” in *Proc. European Conf. Computer Vision (ECCV)*, 2024.